

Legendre–Rvachev Biorthogonal Closure

**Exact finite shifted-up synthesis, the blockwise self-loop,
dual translated moments, rational spectral sum rules, and log-square decay**

Research report for Vladimir Reshetnikov’s ProveIt project

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Abstract

Let $u = u_P$ be Rvachev’s even C^∞ probability density supported on $[-1, 1]$, and let P_ℓ be the Legendre polynomial normalized by $P_\ell(1) = 1$. The repository already proves exact reproduction of every polynomial by a finite local sum of shifted or shifted-and-dilated copies of u , and it proves an absolutely and uniformly convergent Fourier–Legendre series

$$u(x) = \sum_{n \geq 0} u_n P_{2n}(x).$$

This report specializes the first theorem to the Legendre basis, substitutes it into the second theorem, and then develops the additional structure exposed by that closure.

Put

$$M_u(z) = \int_{-1}^1 e^{zx} u(x) dx = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}, \quad \mathcal{D}_u = M_u(D)^{-1}, \quad Q_\ell = \mathcal{D}_u P_\ell.$$

For any integer M with $\nu_2(M) \geq \ell$,

$$P_\ell(x) = \frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) u(x - k/M), \quad -1 \leq x \leq 1.$$

The deconvolved Legendre polynomial has the explicit Jacobi form

$$Q_\ell(x) = \sum_{r=0}^{\lfloor \ell/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} \frac{(\ell + 2r)!}{2^{2r} \ell!} P_{\ell-2r}^{(2r, 2r)}(x), \quad \frac{1}{M_u(z)} = \sum_{r \geq 0} \gamma_r \frac{z^r}{r!}.$$

Choosing $M = 4^n$ for P_{2n} closes the requested loop:

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n).$$

The outer series is absolutely and uniformly convergent only when every finite inner sum is retained as a cancellation block; the adjacent repository report proves that a flattened atom series is not absolute or unconditional.

The principal new structure here is a dual translated-moment kernel

$$\Lambda_m(c) = \frac{2m+1}{2} \int_{-1}^1 u(x-c) P_m(x) dx.$$

It is compactly supported and smooth, satisfies $\Lambda_{2n}(0) = u_n$, and has the spherical-Bessel representation

$$\Lambda_m(c) = (2m+1)(-i)^m \int_{\mathbb{R}} \widehat{u}(\xi) j_m(2\pi\xi) e^{2\pi i \xi c} d\xi.$$

The finite Legendre synthesis and this dual analysis obey exact dyadic biorthogonality:

$$\frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) \Lambda_m(k/M) = \delta_{m\ell}.$$

Expanding every shifted atom back into local Legendre polynomials gives a second closure. It produces explicit rational coefficients $T_{m\ell r}^{(M)}$ for which

$$\delta_{m\ell} = \sum_{r \geq 0} u_r T_{m\ell r}^{(M)}.$$

These are infinite rational spectral identities with exact parity blocks. Endpoint flatness supplies a separate hierarchy of exact cancellation laws,

$$\sum_{r \geq \lceil s/2 \rceil} u_r \frac{(2r+s)!}{2^s s! (2r-s)!} = 0 \quad (s \geq 0).$$

Finally, the sharp real-axis sinc-product envelope from the repository, when combined with the Bessel representation, yields the rigorous coefficient bound

$$|u_n| \leq C \frac{n^\sigma}{\log n} \exp\left(-\frac{(\log n)^2}{2 \log 2}\right)$$

for an explicit finite σ . Exact arithmetic through $n = 80$ suggests the new conjecture $\nu_2(u_n) = -1$ for every n , while moments through μ_{160} suggest $\nu_2(\mu_{2m} - 1) = 3 + \nu_2(m)$. All numerical and symbolic experiments are reproducible from the accompanying commented Python program.

Status and novelty discipline

The Fourier product, probability model, moment and cumulant formulae, exact polynomial synthesis, minimal $d + 2$ endpoint dictionary, canonical Fourier–Legendre expansion, reciprocal-MGF pole asymptotics, and the non-flattening theorem are imported from the current ProveIt corpus. The direct formula for P_ℓ is the requested basis specialization of the existing polynomial theorem. Claims described as “new here” are absent from the canonical live exposition, the consolidated polynomial-synthesis volume, and the directly adjacent Lagrange-loop report after a recursive inventory and targeted full-text search at repository snapshot 73c85cb0023615311fdc4a5ff4951767bddeff34. This is a corpus-relative novelty statement, not an assertion of worldwide priority. Every conjecture is explicitly labeled; numerical evidence is never used in place of proof.

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1 Corpus audit and orientation

1.1 The two transformations being composed

The requested loop is a composition of two exact but analytically very different maps.

First, the up-function generates polynomials locally. For a polynomial p , moment deconvolution produces another polynomial $\mathcal{D}_u p$; samples of that polynomial on a sufficiently oversampled dyadic lattice are coefficients of a finite shifted-up representation of p on a compact interval. This is a Strang–Fix phenomenon whose exact order is controlled by the integer-zero multiplicities

$$\text{ord}_{\xi=k} \widehat{u}(\xi) = 1 + \nu_2(k) \quad (k \in \mathbb{Z} \setminus \{0\}).$$

Second, u itself has a canonical orthogonal expansion in the even Legendre basis. Substituting the finite representation of each P_{2n} gives a self-representation of u . The apparent circularity is not vacuous: the two legs use different coordinate systems and different notions of convergence. Moment deconvolution grows factorially with degree, whereas the outer Legendre coefficients decay superalgebraically. Their product is held together by large finite cancellations.

1.2 Sources used

The recursive corpus inventory records the canonical exposition, archived predecessors, the semi-formalized frontier collection, corrected source papers, and live additions. Mathematically duplicate snapshots were audited through the repository’s own provenance hashes rather than treated as independent sources. The report relies especially on:

- the canonical volume *The Fabius Function and Rvachev’s Up-Function* [13];
- the consolidated volume *Exact Polynomial Synthesis from Rvachev Up-Atoms* [14];
- the immediately adjacent report *Closing the Lagrange–Legendre–Rvachev Loop* [15];
- the historical and analytic sources of Fabius, Rvachev, and Arias de Reyna [1, 2, 3, 4, 5, 6].

1.3 Imported and new results at a glance

Status	Result
Imported	Fourier sinc product, exact integer zero orders, probability law, MGF, rational moments, Bernoulli cumulants, and rapid Fourier decay.
Imported	Exact polynomial synthesis on an interval, the minimal $d + 2$ local end-point dictionary, and the dyadic q -integer/Thue–Morse annihilator.
Requested specialization	Explicit finite shifted-up representation of every Legendre polynomial, including parity reduction and low-degree examples.
New here, corpus-relative	Jacobi-polynomial formula for the entire Rvachev-deconvolved Legendre family Q_ℓ .
Imported/requested composition	Exact blockwise Legendre–up self-loop and fixed-scale partial-sum loop.
Imported from the adjacent report	Factorial asymptotic of $Q_{2n}(0)$ and the theorem that the atom loop cannot be flattened absolutely or unconditionally.
New here, corpus-relative	Dual translated Legendre kernel Λ_m , its Bessel transform, support/parity/rationality, and exact dyadic biorthogonality.
New here, corpus-relative	Reverse local-Legendre loop, clipped affine connection coefficients, and the rational spectral identities $\delta_{m\ell} = \sum_r u_r T_{m\ell r}^{(M)}$.

Status	Result
New here, corpus-relative	Endpoint-flatness coefficient sum rules and finite boundary-jet identities tying the Legendre loop to dyadic derivative cells and Thue–Morse signs.
New here, corpus-relative	A rigorous log-square upper bound on $ u_n $ transferred from the sharp sinc-product Fourier envelope through spherical Bessel functions.
New conjectures	Half-integrality of all u_n , a sharp moment congruence, and dyadically modulated sharp asymptotics for the Legendre coefficients.

2 Foundations and notation

2.1 The up density, Fourier transform, and probability model

Let $u = u_p$ be the unique even nonnegative C^∞ function supported on $[-1, 1]$, normalized by

$$\int_{-1}^1 u(x) dx = 1, \quad u(0) = 1.$$

We use the repository’s Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx.$$

Then

$$\widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}(\xi/2^j), \quad \text{sinc}_{\pi}(t) = \frac{\sin(\pi t)}{\pi t}. \quad (2.1)$$

The zero at a nonzero integer k has exact order $1 + \nu_2(k)$. Equivalently, if V_j are independent uniform random variables on $[-1, 1]$, then

$$Z = \sum_{j \geq 1} 2^{-j} V_j \quad (2.2)$$

has density u . The distributional fixed point is

$$Z \stackrel{d}{=} \frac{V + Z'}{2}, \quad (2.3)$$

where V is uniform on $[-1, 1]$ and Z' is an independent copy of Z .

2.2 Moment generating function, moments, and cumulants

The moment generating function is

$$M_u(z) = \mathbb{E} e^{zZ} = \int_{-1}^1 e^{zx} u(x) dx = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}. \quad (2.4)$$

Write

$$\mu_r = \mathbb{E} Z^r, \quad M_u(z) = \sum_{r \geq 0} \mu_r \frac{z^r}{r!}.$$

All odd moments vanish. From (2.3), for $m \geq 1$,

$$(4^m - 1)\mu_{2m} = \sum_{r=1}^m \binom{2m}{2r} \frac{\mu_{2m-2r}}{2r+1}. \quad (2.5)$$

Thus all moments are rational. The first values are

$$\mu_0 = 1, \quad \mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}.$$

The cumulants are particularly simple:

$$\log M_u(z) = \sum_{r \geq 1} \kappa_r \frac{z^r}{r!}, \quad \kappa_{2r} = \frac{B_{2r}}{2r(1 - 4^{-r})}, \quad \kappa_{2r+1} = 0, \quad (2.6)$$

where B_{2r} are Bernoulli numbers.

2.3 Moment deconvolution and its Bell–Bernoulli coefficients

Define reciprocal coefficients by

$$\frac{1}{M_u(z)} = \sum_{r=0}^{\infty} \gamma_r \frac{z^r}{r!}. \quad (2.7)$$

Again, $\gamma_{2r+1} = 0$. On polynomials, the formal differential operator

$$\mathcal{D}_u := M_u(D)^{-1} = \sum_{r \geq 0} \gamma_r \frac{D^r}{r!} \quad (2.8)$$

terminates exactly. The reciprocal coefficients can be written with complete exponential Bell polynomials:

$$\gamma_n = Y_n(0, -\kappa_2, 0, -\kappa_4, \dots, -\kappa_n). \quad (2.9)$$

Consequently every exact coefficient below belongs to the algebra generated by Bernoulli numbers and rational node coordinates. The first even values are

$$\gamma_0 = 1, \quad \gamma_2 = -\frac{1}{9}, \quad \gamma_4 = \frac{31}{675}, \quad \gamma_6 = -\frac{2347}{59535}, \quad \gamma_8 = \frac{1904369}{32531625}. \quad (2.10)$$

2.4 Legendre and spherical-Bessel normalization

The Legendre polynomials satisfy

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}, \quad \|P_n\|_{L^\infty[-1,1]} = 1. \quad (2.11)$$

Their Jacobi derivative formula is

$$\frac{d^s}{dx^s} P_n(x) = 2^{-s} \frac{(n+s)!}{n!} P_{n-s}^{(s,s)}(x), \quad 0 \leq s \leq n. \quad (2.12)$$

At the endpoint,

$$P_n^{(s)}(1) = \frac{(n+s)!}{2^s s! (n-s)!}. \quad (2.13)$$

Finally, with j_n denoting the spherical Bessel function,

$$\int_{-1}^1 P_n(x) e^{-itx} dx = 2(-i)^n j_n(t). \quad (2.14)$$

These normalizations fix all signs in the dual kernel developed later.

3 Exact finite shifted-up synthesis of Legendre polynomials

3.1 The imported interval synthesis theorem

The polynomial-synthesis volume proves the following physical-scale form.

Theorem 3.1 (Exact unit-radius polynomial synthesis; imported). *Let p be a polynomial of degree at most d , and let M be a positive integer satisfying $\nu_2(M) \geq d$. Then, for every $x \in [-1, 1]$,*

$$p(x) = \frac{1}{M} \sum_{|k| < 2M} (\mathcal{D}_u p)(k/M) u(x - k/M). \quad (3.1)$$

The sum is finite and contains $4M - 1$ literal translates of the unscaled function u .

Repository proof architecture. Use spacing $h = M^{-1}$ and radius-to-spacing ratio M , so the physical atom radius is one. Poisson summation separates the zero-frequency polynomial moment from aliases at nonzero integers. The exact zero multiplicity $1 + \nu_2(k)$ annihilates every alias derivative through degree d , while $M_u(D)^{-1}$ reverses the zero-frequency convolution by the up law. On $[-1, 1]$, the formal endpoint centers $k = \pm 2M$ contribute only values at or outside the support boundary and vanish, leaving (3.1). $\square$

The minimal admissible power-of-two choice is $M = 2^d$. Larger M with the same or greater two-adic valuation gives the same reproduction order on a finer lattice.

3.2 Legendre specialization

Definition 3.2 (Rvachev-deconvolved Legendre family). For $\ell \geq 0$, define

$$Q_\ell(x) := \mathcal{D}_u P_\ell(x). \quad (3.2)$$

Theorem 3.3 (Finite shifted-up representation of P_ℓ). *Let M be a positive integer with $\nu_2(M) \geq \ell$. Then*

$$P_\ell(x) = \frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) u(x - k/M), \quad -1 \leq x \leq 1. \quad (3.3)$$

In particular, $M = 2^\ell$ gives a canonical literal-shift representation with $2^{\ell+2} - 1$ displayed atoms.

Proof. Apply theorem 3.1 to $p = P_\ell$. The definition (3.2) is exactly the required deconvolution. $\square$

This is the requested specialization of the arbitrary-polynomial result. No collocation system or numerical fitting is involved: the coefficient at the center k/M is an exact sample of one explicitly known degree- ℓ polynomial.

3.3 A Jacobi-polynomial formula for Q_ℓ

The arbitrary-polynomial theorem alone does not exploit the special geometry of the Legendre basis. The derivative formula (2.12) makes the specialization explicit.

Theorem 3.4 (Jacobi form of the deconvolved Legendre polynomial; new here). *For every $\ell \geq 0$,*

$$Q_\ell(x) = \sum_{r=0}^{\lfloor \ell/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} \frac{(\ell + 2r)!}{2^{2r} \ell!} P_{\ell-2r}^{(2r, 2r)}(x). \quad (3.4)$$

Consequently $Q_\ell \in \mathbb{Q}[x]$ and

$$Q_\ell(-x) = (-1)^\ell Q_\ell(x). \quad (3.5)$$

Proof. Insert (2.12) with $s = 2r$ into the terminating operator (2.8). Odd reciprocal coefficients vanish. The Jacobi polynomial $P_{\ell-2r}^{(2r,2r)}$ has parity $(-1)^{\ell-2r} = (-1)^\ell$, proving (3.5). Rationality follows from rationality of γ_{2r} and the standard rational coefficients of Jacobi polynomials with integral parameters. $\square$

Remark 3.5 (Bell–Bernoulli content). Combining (3.4), (2.9), and (2.6) expresses every shifted-up coefficient in (3.3) as a finite combination of Jacobi values, complete Bell polynomials, and Bernoulli numbers. Thus the Bell–Bernoulli structure is the exact inverse-smoothing mechanism, not merely a change of notation.

3.4 Parity-reduced atom blocks

The symmetry of Q_ℓ and u halves the independent coefficient data.

Corollary 3.6 (Even and odd paired forms). *If ℓ is even, then*

$$P_\ell(x) = \frac{Q_\ell(0)}{M} u(x) + \frac{1}{M} \sum_{k=1}^{2M-1} Q_\ell(k/M) [u(x - k/M) + u(x + k/M)]. \quad (3.6)$$

If ℓ is odd, then

$$P_\ell(x) = \frac{1}{M} \sum_{k=1}^{2M-1} Q_\ell(k/M) [u(x - k/M) - u(x + k/M)]. \quad (3.7)$$

Proof. Pair the terms indexed by k and $-k$ in (3.3) and use (3.5). $\square$

3.5 Low-degree polynomials and exact examples

The first deconvolved Legendre polynomials are:

$$\begin{aligned} Q_0(x) &= 1, \\ Q_1(x) &= x, \\ Q_2(x) &= \frac{3}{2}x^2 - \frac{2}{3}, \\ Q_3(x) &= \frac{5}{2}x^3 - \frac{7}{3}x, \\ Q_4(x) &= \frac{35}{8}x^4 - \frac{20}{3}x^2 + \frac{134}{135}, \\ Q_5(x) &= \frac{63}{8}x^5 - \frac{35}{2}x^3 + \frac{33}{5}x, \\ Q_6(x) &= \frac{231}{16}x^6 - \frac{175}{4}x^4 + \frac{889}{30}x^2 - \frac{1426}{567}. \end{aligned}$$

For $\ell = 0$, $M = 1$, [theorem 3.3](#) is the partition of unity

$$1 = u(x+1) + u(x) + u(x-1), \quad -1 \leq x \leq 1. \quad (3.8)$$

For $\ell = 1$, $M = 2$,

$$x = \frac{1}{4} \sum_{k=-3}^3 k u(x - k/2). \quad (3.9)$$

For $\ell = 2$, $M = 4$,

$$P_2(x) = \sum_{k=-7}^7 \frac{9k^2 - 64}{384} u(x - k/4). \quad (3.10)$$

These identities hold on the whole closed interval, including endpoints.

3.6 The compressed $\ell + 2$ -atom representation

Literal unit-radius shifts make the formula transparent but require exponentially many displayed centers. The repository’s local dimension theorem compresses any degree- ℓ polynomial to $\ell + 2$ atoms at a large common dilation.

Put $M = 2^\ell$, $t = (x + 1)/2$, and

$$R_\ell(t) = P_\ell(2t - 1), \quad \tilde{Q}_\ell(t) = M_u(MD_t)^{-1}R_\ell(t). \quad (3.11)$$

Let

$$A_\ell(z) = \prod_{j=1}^{\ell} [2^j]_z, \quad [N]_z = 1 + z + \cdots + z^{N-1}. \quad (3.12)$$

Its degree is $2M - \ell - 2$. The complete local alias nullspace is generated by this q -integer product; after subtracting the unique null sequence matching the left coefficients, only the centers

$$M - \ell - 1 \leq k \leq M \quad (3.13)$$

remain.

Theorem 3.7 (Compressed Legendre synthesis; imported specialization). *There are unique rational coefficients $b_{\ell,k}$, with k in (3.13), such that*

$$P_\ell(x) = \frac{1}{M} \sum_{k=M-\ell-1}^M b_{\ell,k} u\left(\frac{x+1-2k}{2M}\right), \quad -1 \leq x \leq 1. \quad (3.14)$$

Thus only $\ell + 2$ shifted-and-dilated up-atoms are needed.

Algorithm 3.8 (Exact coefficient compression). Write $A_\ell(z) = \sum_{r=0}^R a_r z^r$ with $a_R = 1$. On the full active integer interval $-M + 1 \leq k \leq M$, initialize a null sequence by matching $n_k = \tilde{Q}_\ell(k)$ on the first R positions and extend it by

$$n_{s+R} = - \sum_{r=0}^{R-1} a_r n_{s+r}. \quad (3.15)$$

Then

$$b_{\ell,k} = \tilde{Q}_\ell(k) - n_k, \quad M - \ell - 1 \leq k \leq M. \quad (3.16)$$

All operations are exact in $\mathbb{Q}$.

The price of linear atom count is a dilation of order 2^ℓ and substantial coefficient cancellation. The literal and compressed formulas therefore occupy opposite ends of a scale–cardinality–conditioning tradeoff.

4 The canonical Fourier–Legendre expansion and the requested loop

4.1 Exact Legendre coefficients

The canonical repository expansion is

$$u(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x), \quad u_n = \frac{4n+1}{2} \int_{-1}^1 u(x) P_{2n}(x) dx. \quad (4.1)$$

It converges absolutely and uniformly on $[-1, 1]$. Expanding P_{2n} in monomials gives the exact moment formula

$$u_n = \frac{4n+1}{2^{2n+1}} \sum_{s=0}^n (-1)^{n-s} \frac{(2n+2s)!}{(n-s)!(n+s)!(2s)!} \mu_{2s}. \quad (4.2)$$

Thus $u_n \in \mathbb{Q}$.

n	exact u_n	decimal value	$v_2(u_n)$
0	$\frac{1}{2}$	0.5	−1
1	$-\frac{5}{6}$	−0.8333333333	−1
2	$\frac{11}{30}$	0.3666666667	−1
3	$\frac{143}{5670}$	0.02522045855	−1
4	$-\frac{7291}{85050}$	−0.0857260435	−1
5	$\frac{2485890}{9726162257}$	0.03987988205	−1
6	$-\frac{213774110550}{27288832157947}$	−0.02678604178	−1
7	$\frac{1167420417713550}{144048691033129139}$	0.02337532541	−1
8	$-\frac{11591954102251136250}{245262858648906702103531}$	−0.01242660985	−1
9	$\frac{100278737599471847129463750}{190920333865973179103798951}$	0.002445811191	−1
10	$\frac{82729958519564273881807593750}{45956358119729097218525273579033}$	0.002307753289	−1
11	$-\frac{23132967547232266175685815725893750}{23132967547232266175685815725893750}$	−0.001986617498	−1

4.2 Termwise finite closure

For P_{2n} , take the minimal power-of-two ratio

$$M_n = 2^{2n} = 4^n. \quad (4.3)$$

Then [theorem 3.3](#) gives

$$P_{2n}(x) = \frac{1}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n). \quad (4.4)$$

Theorem 4.1 (Blockwise Legendre–Rvachev self-loop). *For every $x \in [-1, 1]$,*

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n). \quad (4.5)$$

The outer series is absolutely and uniformly convergent when each finite inner sum is retained as one block.

Proof. Substitute (4.4) into (4.1). The n th finite inner sum is exactly $P_{2n}(x)$, so its full block is $u_n P_{2n}(x)$. Since $\|P_{2n}\|_{\infty} = 1$ and the canonical expansion is absolutely uniformly convergent,

$$\sum_{n \geq 0} \|u_n P_{2n}\|_{\infty} = \sum_{n \geq 0} |u_n| < \infty.$$

□

Equation (4.5) is the requested closed loop. It is an exact series of finite self-reconstruction operators; it is not a tautological pointwise insertion because the atom coefficients are explicit rational samples of the deconvolved Legendre polynomials.

4.3 One lattice for an entire partial sum

Define

$$S_N(x) = \sum_{n=0}^N u_n P_{2n}(x), \quad C_N(x) = \mathcal{D}_u S_N(x) = \sum_{n=0}^N u_n Q_{2n}(x). \quad (4.6)$$

Since $\deg S_N = 2N$, the single ratio $M = 4^N$ reproduces the whole partial sum.

Corollary 4.2 (Fixed-scale finite self-approximants). *For $x \in [-1, 1]$,*

$$S_N(x) = \frac{1}{4^N} \sum_{|k| < 2 \cdot 4^N} C_N(k/4^N) u(x - k/4^N). \quad (4.7)$$

Every coefficient is rational, and the right side converges uniformly to u as $N \rightarrow \infty$.

Applying the compressed dictionary to S_N reduces the atom count to $2N + 2$, again at the price of dilation by 4^N and much larger cancellation.

4.4 Why the loop must remain blockwise

The adjacent Lagrange-loop report analyzes the repeated central copy $u(x)$ in (4.5). With

$$C_* = M_u(\pi i) = \prod_{j \geq 1} \frac{\sin(\pi/2^j)}{\pi/2^j}, \quad (4.8)$$

it proves

$$Q_{2n}(0) = \frac{2(-1)^n}{C_*} \frac{(4n)!}{(2n)!(4\pi)^{2n}} \left(1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2}) \right). \quad (4.9)$$

In particular,

$$|Q_{2n}(0)|^{1/(2n)} \sim \frac{2n}{\pi e}. \quad (4.10)$$

The same report proves that

$$\beta_n := \frac{u_n Q_{2n}(0)}{4^n} \quad (4.11)$$

is unbounded in absolute value. Otherwise u_n would decay supergeometrically, forcing an analytic continuation of u across its flat support endpoints, an impossibility for a nonzero compactly supported C^∞ function. Therefore the double atom series cannot be flattened into an absolutely or unconditionally convergent scalar series. Increasingly large signed coefficients cancel inside the n th block to leave the small function $u_n P_{2n}$.

Warning 4.3 (Legal and illegal rearrangements). Finite regrouping inside one block is always legitimate, and the outer block series may be rearranged absolutely. Grouping all copies with the same center across infinitely many degrees is not justified in general; the central coefficient sequence alone is unbounded.

5 A dual translated Legendre kernel

The finite synthesis coefficients $Q_\ell(k/M)$ analyze a polynomial before placing up-atoms. To close the transform in the opposite direction, we need the Legendre coefficients of a shifted up-atom.

Definition 5.1 (Translated Legendre analysis kernel). For $m \geq 0$ and $c \in \mathbb{R}$, define

$$\Lambda_m(c) := \frac{2m+1}{2} \int_{-1}^1 u(x-c) P_m(x) dx. \quad (5.1)$$

5.1 Elementary properties

Proposition 5.2 (Support, smoothness, probability form, and parity; new here). *For every $m \geq 0$:*

1. $\Lambda_m \in C_c^\infty(\mathbb{R})$ and $\text{supp } \Lambda_m \subseteq [-2, 2]$;
2. if Z has density u , then

$$\Lambda_m(c) = \frac{2m+1}{2} \mathbb{E}[P_m(c+Z) \mathbf{1}_{\{|c+Z| \leq 1\}}]; \quad (5.2)$$

- 3.

$$\Lambda_m(-c) = (-1)^m \Lambda_m(c); \quad (5.3)$$

4. at the origin,

$$\Lambda_{2n}(0) = u_n, \quad \Lambda_{2n+1}(0) = 0. \quad (5.4)$$

Proof. Let $\chi_m(x) = P_m(x)\mathbf{1}_{[-1,1]}(x)$. Since u is even,

$$\Lambda_m(c) = \frac{2m+1}{2}(u * \chi_m)(c).$$

Convolution of a compactly supported C^∞ function with a compactly supported integrable function is C^∞ ; supports add, giving $[-2, 2]$. Changing variables $x = c + z$ gives (5.2). Reflection $x \mapsto -x$, together with evenness of u and parity of P_m , proves (5.3). Finally, (5.4) is the definition of the canonical Legendre coefficient, with odd terms zero by symmetry. $\square$

Thus the original coefficients u_n are merely the central samples of a much larger family of translated Legendre moments.

5.2 Spherical-Bessel transform

Theorem 5.3 (Fourier–Bessel representation of Λ_m ; new here). *For all $m \geq 0$ and $c \in \mathbb{R}$,*

$$\Lambda_m(c) = (2m+1)(-i)^m \int_{\mathbb{R}} \widehat{u}(\xi) j_m(2\pi\xi) e^{2\pi i \xi c} d\xi. \quad (5.5)$$

In particular,

$$u_n = (4n+1)(-1)^n \int_{\mathbb{R}} \widehat{u}(\xi) j_{2n}(2\pi\xi) d\xi. \quad (5.6)$$

Proof. The Fourier transform of χ_m is, by (2.14),

$$\widehat{\chi}_m(\xi) = 2(-i)^m j_m(2\pi\xi).$$

Since $\Lambda_m = (2m+1)(u * \chi_m)/2$, Fourier inversion gives (5.5). Set $c = 0$ and $m = 2n$ to obtain (5.6). $\square$

Equation (5.6) is the analytic bridge between the repository’s sharp sinc-product Fourier envelope and the Legendre coefficients. It also suggests that dyadic Fourier-lobe structure should survive as a modulation in u_n .

5.3 Exact rationality at dyadic shifts

Proposition 5.4 (Dyadic rationality of the dual samples; new here). *If c is dyadic rational, then $\Lambda_m(c) \in \mathbb{Q}$ for every m .*

Proof. After $x = c + y$,

$$\Lambda_m(c) = \frac{2m+1}{2} \int_{[-1,1] \cap [-1-c, 1-c]} u(y) P_m(y+c) dy.$$

The endpoints are dyadic. Expand $P_m(y+c)$ into powers of y . The repository’s exact dyadic primitive calculus expresses every truncated moment $\int_a^b y^r u(y) dy$ at dyadic endpoints in rational Fabius/up values and rational polynomial factors. Hence the displayed finite linear combination is rational. $\square$

The accompanying code does not need to evaluate u to compute the later connection coefficients; it integrates the corresponding clipped Legendre polynomials exactly in $\mathbb{Q}$.

6 Exact dyadic spectral biorthogonality

6.1 The scalar identity

Theorem 6.1 (Legendre–up dyadic biorthogonality; new here). *Let $d \geq 0$, $M = 2^d$, $0 \leq \ell \leq d$, and $m \geq 0$. Then*

$$\boxed{\frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) \Lambda_m(k/M) = \delta_{m\ell}.} \quad (6.1)$$

More generally, the same identity holds for any positive integer M with $\nu_2(M) \geq \ell$.

Proof. Use the admissible common-scale representation

$$P_\ell(x) = \frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) u(x - k/M).$$

Multiply by $(2m+1)P_m(x)/2$ and integrate over $[-1, 1]$. Orthogonality gives $\delta_{m\ell}$ on the left, while the integral of the shifted atom is exactly $\Lambda_m(k/M)$ by [definition 5.1](#). $\square$

For each fixed ℓ , (6.1) is an infinite family of finite identities indexed by m . The cases $m > \ell$ say that the finite coefficient row annihilates all higher translated Legendre moments exactly.

6.2 Parity blocks

From (3.5) and (5.3), the summand in (6.1) changes by $(-1)^{m+\ell}$ under $k \mapsto -k$. Therefore opposite parities cancel term by term. For equal parity,

$$\delta_{m\ell} = \frac{Q_\ell(0)\Lambda_m(0)}{M} + \frac{2}{M} \sum_{k=1}^{2M-1} Q_\ell(k/M) \Lambda_m(k/M), \quad (6.2)$$

where the central term vanishes automatically in the odd–odd block.

6.3 Finite analysis and synthesis matrices

Let

$$\mathcal{K}_M = \{k \in \mathbb{Z} : |k| < 2M\}.$$

For $0 \leq m, \ell \leq d$, define

$$A_{k\ell}^{(d)} = \frac{1}{M} Q_\ell(k/M), \quad L_{mk}^{(d)} = \Lambda_m(k/M). \quad (6.3)$$

Then $A^{(d)}$ is a $(4M-1) \times (d+1)$ synthesis matrix and $L^{(d)}$ is a $(d+1) \times (4M-1)$ analysis matrix.

Corollary 6.2 (Exact spectral right inverse and oblique projector). *The matrices in (6.3) satisfy*

$$\boxed{L^{(d)} A^{(d)} = I_{d+1}.} \quad (6.4)$$

Consequently

$$\Pi^{(d)} := A^{(d)} L^{(d)}, \quad (\Pi^{(d)})^2 = \Pi^{(d)}, \quad \text{rank } \Pi^{(d)} = \text{tr } \Pi^{(d)} = d+1. \quad (6.5)$$

The even and odd Legendre modes form independent parity blocks.

This is the spectral counterpart of the nodal right inverse in the adjacent Lagrange report. There, analysis consists of point evaluation at interpolation nodes. Here, analysis consists of exact Legendre moments of shifted atoms.

Corollary 6.3 (Trace and Cauchy–Binet identities). *One has*

$$\frac{1}{M} \sum_{k \in \mathcal{K}_M} \sum_{\ell=0}^d Q_\ell(k/M) \Lambda_\ell(k/M) = d + 1, \quad (6.6)$$

and

$$1 = \sum_{\substack{S \subset \mathcal{K}_M \\ |S|=d+1}} \det L_{:,S}^{(d)} \det A_{S,:}^{(d)}. \quad (6.7)$$

Proof. Take the trace and determinant, respectively, of $L^{(d)} A^{(d)} = I$; the determinant expansion is Cauchy–Binet. $\square$

7 The reverse local-Legendre loop

The requested self-loop substitutes an up-atom representation into a Legendre series. We can also expand each shifted atom back into Legendre polynomials in its own local coordinate. The support clipping is essential.

7.1 Shifted local expansions

For $c \in [-2, 2]$, define the overlap interval

$$I(c) = [-1, 1] \cap [c - 1, c + 1]. \quad (7.1)$$

The canonical series implies, uniformly for real x ,

$$u(x - c) = \mathbf{1}_{I(c)}(x) \sum_{r=0}^{\infty} u_r P_{2r}(x - c). \quad (7.2)$$

Indeed, inside the support this is the canonical expansion in the coordinate $x - c$, and outside both sides vanish. Absolute uniform convergence survives multiplication by the indicator because $\sum_r |u_r| < \infty$ and $|P_{2r}| \leq 1$ on $[-1, 1]$.

For admissible M , define the piecewise-polynomial block

$$H_{\ell r}^{(M)}(x) = \frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) \mathbf{1}_{I(k/M)}(x) P_{2r}(x - k/M). \quad (7.3)$$

Its knots belong to the dyadic grid generated by the endpoints $k/M \pm 1$.

Theorem 7.1 (Reverse local-Legendre closure; new here). *If $\nu_2(M) \geq \ell$, then on $[-1, 1]$,*

$$\boxed{P_\ell(x) = \sum_{r=0}^{\infty} u_r H_{\ell r}^{(M)}(x).} \quad (7.4)$$

The series converges absolutely and uniformly.

Proof. Insert (7.2) into the finite synthesis formula (3.3). Since the outer atom sum is finite and each local Legendre series is absolutely uniform, interchange of the two sums is legitimate. The coefficient of u_r is exactly (7.3). $\square$

Although each $H_{\ell r}^{(M)}$ is only piecewise polynomial, the infinite weighted sum is the globally smooth polynomial P_ℓ . All jumps of all piecewise derivatives cancel. Section 8 identifies the endpoint sum rules responsible for this phenomenon.

7.2 Clipped affine Legendre connection coefficients

Project the local expansion against a global Legendre mode. Define

$$C_{m,r}(c) = \frac{2m+1}{2} \int_{I(c)} P_m(x) P_{2r}(x-c) dx. \quad (7.5)$$

For rational c , this is rational because it is the integral of a polynomial over an interval with rational endpoints. Reflection gives

$$C_{m,r}(-c) = (-1)^m C_{m,r}(c). \quad (7.6)$$

Proposition 7.2 (Local connection expansion of the dual kernel; new here). *For every $m \geq 0$ and $c \in [-2, 2]$,*

$$\Lambda_m(c) = \sum_{r=0}^{\infty} u_r C_{m,r}(c). \quad (7.7)$$

The series converges absolutely and uniformly in c .

Proof. Insert (7.2) into (5.1) and integrate termwise. Uniformity follows from $|P_m| \leq 1$, $|P_{2r}| \leq 1$ on their respective intervals, interval length at most two, and $\sum_r |u_r| < \infty$. $\square$

7.3 Arithmetic spectral closure

For $\nu_2(M) \geq \ell$, define

$$T_{m\ell r}^{(M)} = \frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) C_{m,r}(k/M). \quad (7.8)$$

Every $T_{m\ell r}^{(M)}$ is rational.

Theorem 7.3 (Infinite rational spectral identities; new here). *For every $m, \ell \geq 0$ and every admissible M with $\nu_2(M) \geq \ell$,*

$$\delta_{m\ell} = \sum_{r=0}^{\infty} u_r T_{m\ell r}^{(M)}. \quad (7.9)$$

Moreover,

$$m + \ell \text{ odd} \implies T_{m\ell r}^{(M)} = 0 \text{ for every } r. \quad (7.10)$$

Proof. Substitute (7.7) into the finite biorthogonality identity (6.1). The finite sum over k may be interchanged with the absolutely convergent series over r , giving (7.9). Rationality follows from rationality of $Q_\ell(k/M)$ and $C_{m,r}(k/M)$. Finally, $Q_\ell(-c)C_{m,r}(-c) = (-1)^{m+\ell}Q_\ell(c)C_{m,r}(c)$; opposite parity therefore cancels in symmetric pairs, while the central term vanishes when appropriate. $\square$

This theorem converts an analytic orthogonality relation into infinitely many identities involving only rational numbers. All transcendental content has been isolated in the universal coefficient sequence (u_r) .

7.4 Exact examples

For $(\ell, m) = (1, 1)$ and $M = 2$, the first connection constants are

$$\left(T_{11r}^{(2)} \right)_{r \geq 0} = \left(\frac{15}{8}, -\frac{15}{128}, -\frac{85}{1024}, -\frac{267}{32768}, \frac{11949}{262144}, -\frac{128339}{4194304}, \dots \right), \quad (7.11)$$

and (7.9) says

$$1 = \sum_{r \geq 0} u_r T_{11r}^{(2)}. \quad (7.12)$$

For $(\ell, m) = (2, 2)$, $M = 4$,

$$\left(T_{22r}^{(4)}\right)_{r \geq 0} = \left(\frac{945}{512}, -\frac{29737}{196608}, -\frac{291095}{2097152}, -\frac{28532595}{268435456}, -\frac{400224525}{8589934592}, \dots\right), \quad (7.13)$$

again summing to one after weighting by u_r . In contrast,

$$\left(T_{20r}^{(4)}\right)_{r \geq 0} = \left(\frac{61}{96}, \frac{189}{512}, -\frac{945}{32768}, -\frac{24003}{1048576}, -\frac{1504629}{134217728}, \dots\right) \quad (7.14)$$

has weighted sum zero. Every $T_{21r}^{(4)}$ is exactly zero by parity.

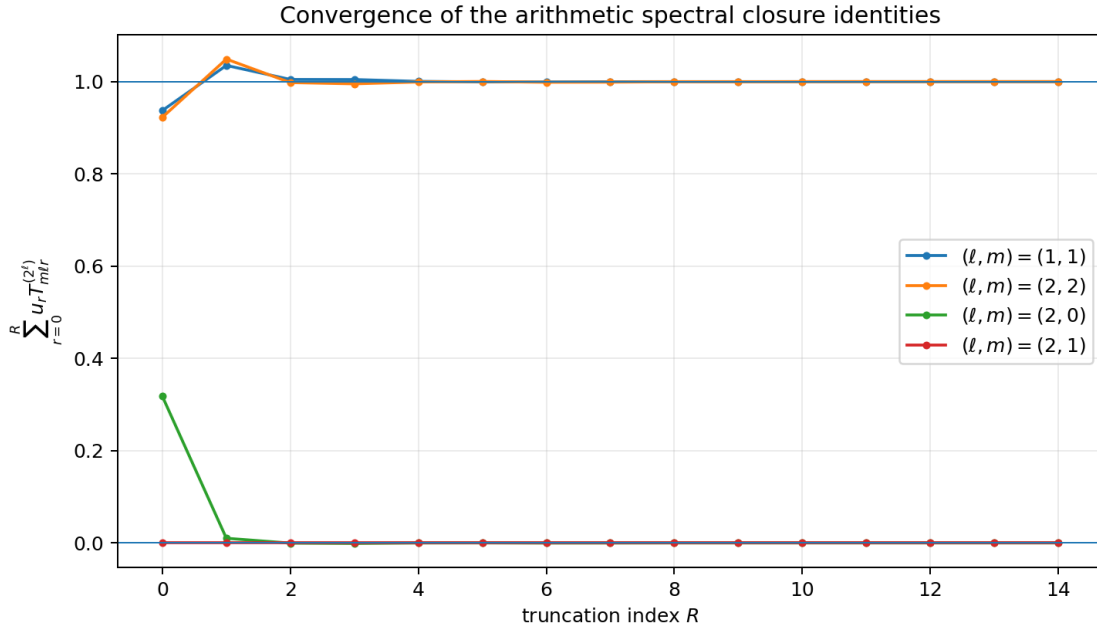


Figure 1: Partial sums of four exact rational spectral identities. Decimal plotting is used only for visualization; the coefficients and all partial sums in the accompanying CSV file are computed in exact rational arithmetic.

8 Endpoint flatness, boundary jets, and Thue–Morse cancellation

8.1 Infinite Legendre coefficient sum rules

The up-function is C^∞ on $\mathbb{R}$ and vanishes outside $[-1, 1]$. Therefore

$$u^{(s)}(1) = u^{(s)}(-1) = 0 \quad (s \geq 0). \quad (8.1)$$

Smoothness implies that Legendre coefficients decay faster than every fixed power, so the canonical series may be differentiated termwise any prescribed finite number of times.

Theorem 8.1 (Endpoint-flatness sum rules; new here). *For every integer $s \geq 0$,*

$$\sum_{r \geq \lceil s/2 \rceil} u_r \frac{(2r+s)!}{2^s s! (2r-s)!} = 0. \quad (8.2)$$

The series converges absolutely.

Proof. Differentiate (4.1) s times, evaluate at $x = 1$, and use (8.1) and (2.13). Terms with $2r < s$ vanish. Absolute convergence follows from superalgebraic decay of u_r and polynomial growth of the endpoint derivative. $\square$

The first three identities are

$$\sum_{r \geq 0} u_r = 0, \quad (8.3)$$

$$\sum_{r \geq 1} r(2r + 1)u_r = 0, \quad (8.4)$$

$$\sum_{r \geq 1} \frac{(2r - 1)(2r)(2r + 1)(2r + 2)}{8} u_r = 0. \quad (8.5)$$

They quantify how a series whose zeroth coefficient is $1/2$ nevertheless vanishes to infinite order at both endpoints.

8.2 Finite boundary-jet identities

The finite synthesis formula itself can be differentiated exactly.

Theorem 8.2 (Boundary jets of the finite atom block; new here). *Let $\nu_2(M) \geq \ell$. For every $s \geq 0$,*

$$\frac{1}{M} \sum_{|k| < 2M} Q_\ell(k/M) u^{(s)}(1 - k/M) = \begin{cases} \frac{(\ell + s)!}{2^s s! (\ell - s)!}, & s \leq \ell, \\ 0, & s > \ell. \end{cases} \quad (8.6)$$

At the left endpoint, the right side is multiplied by $(-1)^{\ell+s}$.

Proof. Differentiate (3.3) s times. The sum is finite, so there is no convergence issue. Evaluate at $x = 1$ and use (2.13); if $s > \ell$, the derivative of the polynomial vanishes. Reflection gives the left endpoint formula. $\square$

These identities are finite counterparts of (8.2). They state that a signed collection of flat compact atoms exactly reproduces every boundary derivative of P_ℓ and annihilates all derivatives above its degree.

8.3 Dyadic derivative cells and Thue–Morse signs

The primary repository proves the exact cell law

$$u^{(s)}\left(-1 + \frac{2j + 1 + y}{2^s}\right) = 2^{\binom{s+1}{2}} (-1)^{\text{wt}_2(j)} u(y), \quad (8.7)$$

for $0 \leq j < 2^s$ and $-1 \leq y \leq 1$. Every argument $1 - k/M$ in (8.6) is dyadic. Locate it in its level- s cell and apply (8.7). The boundary-jet theorem then becomes an exact finite sum of dyadic up-values weighted by:

- samples of the Bell–Bernoulli/Jacobi polynomial Q_ℓ ;
- the scale factor $2^{\binom{s+1}{2}}$;
- a Thue–Morse sign $(-1)^{\text{wt}_2(j)}$.

Since dyadic up-values are rational, this supplies a large family of exact rational Thue–Morse identities. It is the derivative-side counterpart of the q -integer annihilator used to compress the atom dictionary.

9 A log-square bound for the Legendre coefficients

The canonical exposition proves the sharp real-axis Fourier envelope

$$|\widehat{u}(\xi)| \leq C_0 E_{\kappa_\infty}(|\xi|), \quad |\xi| \geq 1, \quad (9.1)$$

where

$$E_\kappa(x) = \exp\left(-\frac{(\log x)^2}{2 \log 2}\right) x^{-\kappa}, \quad \kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2}. \quad (9.2)$$

The coefficient formula (5.6) transfers this estimate to Legendre space.

9.1 A Bessel small-argument bound

Lemma 9.1 (Exponential suppression below the turning point). *There exist constants $C_1 > 0$ and $0 < q < 1$ such that, for every sufficiently large integer m and every real t with $|t| \leq m/2$,*

$$|j_m(t)| \leq C_1 q^m. \quad (9.3)$$

For all $m \geq 0$ and real t ,

$$|j_m(t)| \leq 1. \quad (9.4)$$

Proof. The power series for J_ν gives

$$|J_\nu(t)| \leq \frac{(|t|/2)^\nu}{\Gamma(\nu+1)} \exp\left(\frac{t^2}{4(\nu+1)}\right).$$

Using $j_m(t) = \sqrt{\pi/(2t)} J_{m+1/2}(t)$ yields

$$|j_m(t)| \leq \frac{\sqrt{\pi} |t|^m}{2^{m+1} \Gamma(m+3/2)} \exp\left(\frac{t^2}{4(m+3/2)}\right). \quad (9.5)$$

Stirling's formula, uniformly for $|t| \leq m/2$, makes the right side at most $C_1 q^m$ for some $q < 1$. The global bound follows directly from (2.14) and $|P_m(x)| \leq 1$: $2|j_m(t)| \leq \int_{-1}^1 |P_m(x)| dx \leq 2$. $\square$

9.2 The tail integral

Lemma 9.2 (Explicit log-normal tail bound). *Let $a = \log 2$ and $\kappa > 1$. For all sufficiently large X ,*

$$\int_X^\infty E_\kappa(x) dx \leq \frac{a}{\log X + (\kappa - 1)a} X^{1-\kappa} \exp\left(-\frac{(\log X)^2}{2a}\right). \quad (9.6)$$

Proof. Put $t = \log x$. The integral becomes

$$\int_{\log X}^\infty \exp\left(-\frac{t^2}{2a} - (\kappa - 1)t\right) dt.$$

The exponent's negative logarithm is convex with derivative $t/a + \kappa - 1$. Its tangent at $t_0 = \log X$ is a lower bound, so integration of the resulting exponential majorant gives (9.6). $\square$

9.3 The coefficient theorem

Theorem 9.3 (Log-square decay inherited from the sinc product; new here). *There is a constant $C > 0$ such that, for all sufficiently large n ,*

$$|u_n| \leq C \frac{n^\sigma}{\log n} \exp\left(-\frac{(\log n)^2}{2 \log 2}\right), \quad (9.7)$$

where one admissible explicit exponent is

$$\sigma = 2 - \kappa_\infty + \log_2(2\pi). \quad (9.8)$$

Consequently

$$\liminf_{\substack{n \rightarrow \infty \\ u_n \neq 0}} \frac{-\log |u_n|}{(\log n)^2} \geq \frac{1}{2 \log 2}. \quad (9.9)$$

Proof. Use (5.6) and split the integral at

$$X_n = \frac{n}{2\pi}.$$

On $|\xi| \leq X_n$, the Bessel argument satisfies $|2\pi\xi| \leq n = (2n)/2$, so lemma 9.1 with $m = 2n$ gives an inner contribution $\mathcal{O}(n^2 q^{2n})$ after accounting for interval length and the prefactor $4n + 1$.

On the complement, use $|j_{2n}| \leq 1$ and the Fourier envelope (9.1):

$$|u_n| \leq C_2 n^2 q^{2n} + C_3 n \int_{X_n}^\infty E_{\kappa_\infty}(x) dx.$$

Apply lemma 9.2. Since $\log X_n = \log n - \log(2\pi)$,

$$\begin{aligned} & n X_n^{1-\kappa_\infty} \exp\left(-\frac{(\log X_n)^2}{2 \log 2}\right) \\ &= \mathcal{O}\left(n^{2-\kappa_\infty+\log_2(2\pi)} \exp\left(-\frac{(\log n)^2}{2 \log 2}\right)\right). \end{aligned}$$

The denominator in (9.6) contributes $1/\log n$. The exponentially small inner term is absorbed. Taking logarithms proves (9.9). $\square$

The theorem is stronger than generic “faster than any power” smoothness: it retains the precise quadratic logarithmic scale dictated by the dyadic sinc product. It is not asserted to be asymptotically sharp; the lower-bound problem requires controlling cancellation in the oscillatory Bessel integral.

10 Two-adic arithmetic of moments and Legendre coefficients

10.1 A proved unit theorem for the moments

All rational numbers in this subsection are viewed inside $\mathbb{Q}_2$.

Theorem 10.1 (Every even up-moment is a two-adic unit; new elementary proof). *For every $m \geq 0$,*

$$\nu_2(\mu_{2m}) = 0, \quad \mu_{2m} \equiv 1 \pmod{2}. \quad (10.1)$$

Proof. Induct on m . The case $m = 0$ is clear. In (2.5), every denominator $2r + 1$ and the factor $4^m - 1$ are odd. By induction, each earlier moment is congruent to one modulo two. Therefore the right side is congruent to

$$\sum_{r=1}^m \binom{2m}{2r} = 2^{2m-1} - 1 \equiv 1 \pmod{2}.$$

Division by the odd unit $4^m - 1$ preserves the congruence. $\square$

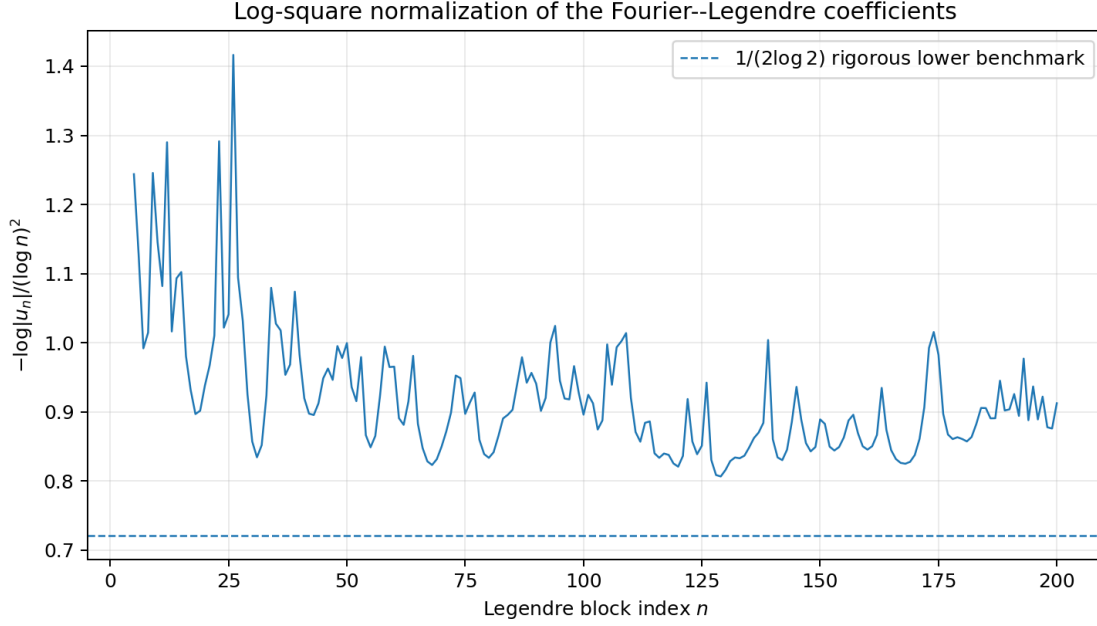


Figure 2: High-precision values of $-\log |u_n|/(\log n)^2$ through $n = 200$. The dashed line is the rigorous benchmark from [theorem 9.3](#). Oscillation remains substantial, consistent with a dyadic modulation inherited from the Fourier lobes.

10.2 Experimental congruence patterns

Exact computation of (2.5) through $m = 80$ gives the stronger pattern

$$\nu_2(\mu_{2m} - 1) = 3 + \nu_2(m). \quad (10.2)$$

Independently, the exact moment formula (4.2) gives

$$\nu_2(u_n) = -1 \quad (0 \leq n \leq 80). \quad (10.3)$$

Thus every tested reduced denominator of u_n is twice an odd integer, and every tested numerator is odd. In particular, none of the first eighty-one coefficients vanishes.

Conjecture 10.2 (Sharp moment congruence). *For every $m \geq 1$,*

$$\boxed{\nu_2(\mu_{2m} - 1) = 3 + \nu_2(m)}. \quad (10.4)$$

Conjecture 10.3 (Half-integrality and nonvanishing of Legendre coefficients). *For every $n \geq 0$,*

$$\boxed{\nu_2(u_n) = -1}. \quad (10.5)$$

Equivalently, $2u_n$ is a two-adic unit. In particular $u_n \neq 0$ for every n .

A proof of [conjecture 10.3](#) cannot simply reduce the monomial formula for P_{2n} modulo two, because its coefficients have large powers of two in their denominators. Promising approaches include a two-adic analysis of the generating function

$$\mathbb{E}(1 - 2tZ + t^2)^{-1/2} = \sum_{n \geq 0} \mathbb{E}[P_n(Z)]t^n,$$

or a filtration that combines [conjecture 10.2](#) with the exact Kummer valuations of the Legendre coefficients.

11 Numerical experiments and independent checks

11.1 Exact arithmetic layer

The accompanying program `experiments.py` uses Python’s `fractions.Fraction` for:

- the moment recurrence (2.5);
- reciprocal-MGF coefficients from exponential-generating-function inversion;
- exact monomial coefficients of P_n and Q_n ;
- exact clipped integrals $C_{m,r}(c)$;
- exact spectral constants $T_{m\ell r}^{(M)}$ and their partial sums;
- two-adic valuations of all reduced rational outputs.

No sampled value of u is involved in these calculations.

11.2 High-precision coefficient layer

The finite moment expression for u_n has severe alternating cancellation. The script evaluates moments and coefficient sums with 2300 decimal digits through $n = 200$. Selected normalized values are

n	20	40	60	80	100	120	140	160	180	200
$-\log u_n /(\log n)^2$	.9381	.9823	.9654	.8336	.8961	.8208	.8604	.8455	.8609	.9125

The rigorous benchmark is $1/(2 \log 2) = 0.7213475204 \dots$. The data show no monotone approach and strong local oscillation.

11.3 Independent Fourier-product synthesis check

To check (3.3) without reusing the symbolic proof, the script evaluates u by inverse FFT of a 42-factor truncation of (2.1) on a period-eight grid with 2^{18} samples. It then reconstructs P_ℓ from the exact rational coefficients. The maximum residuals on 2001 points of $[-1, 1]$ are:

ℓ	0	1	2	3	4	5
max residual	1.22×10^{-15}	8.88×10^{-16}	3.49×10^{-10}	1.73×10^{-9}	5.19×10^{-9}	1.21×10^{-8}

The growth is expected: exact deconvolution coefficients amplify the small FFT and interpolation errors increasingly strongly.

11.4 Reproduction

From the artifact directory, run

```
python experiments.py
```

The default run regenerates all CSV files, exact TeX snippets, and figures. Use

```
python experiments.py --no-plots
```

to omit Matplotlib output. The program requires only NumPy, Matplotlib, mpmath, and the Python standard library.

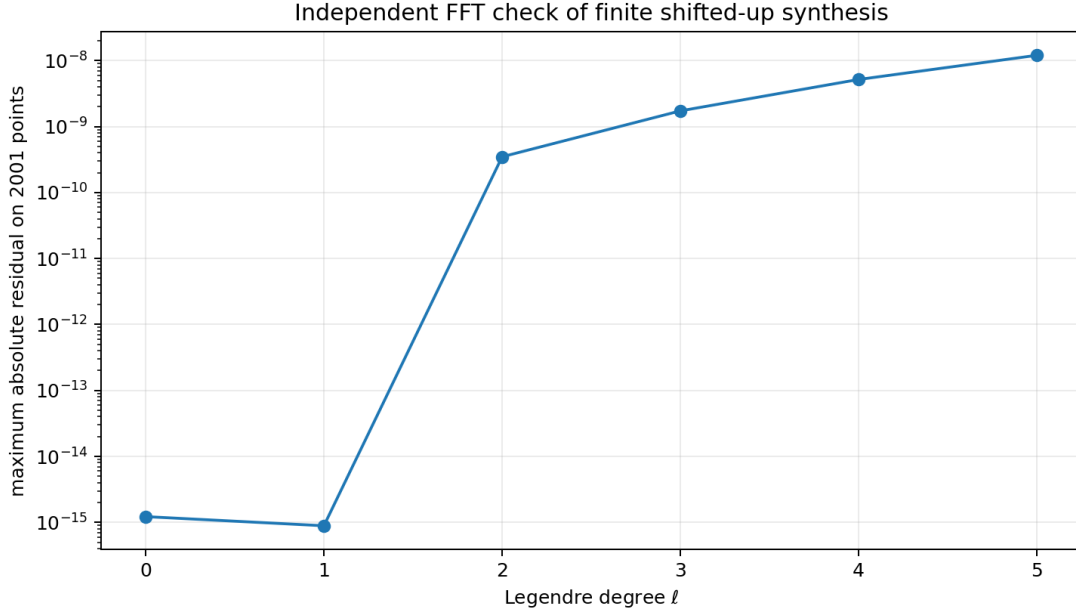


Figure 3: Independent numerical residuals of the exact finite synthesis formula. The identities themselves are symbolic; the graph measures only error in the FFT approximation of u and floating evaluation of a cancellation-prone sum.

12 Frontier conjectures and research directions

12.1 Sharp transfer of the Fourier envelope

Conjecture 12.1 (Sharp log-square constant). *The lower benchmark in (9.9) is sharp:*

$$\liminf_{n \rightarrow \infty} \frac{-\log |u_n|}{(\log n)^2} = \frac{1}{2 \log 2}. \quad (12.1)$$

If conjecture 10.3 holds, no exclusion of zero coefficients is needed.

The spherical Bessel kernel is concentrated near its turning point $2\pi|\xi| \asymp 2n$, so the conjecture predicts that subsequences of Legendre coefficients detect near-extremal Fourier lobes rather than suffering an additional quadratic-log cancellation penalty.

Conjecture 12.2 (Dyadic modulation). *After removing the principal factor $\exp[-(\log n)^2/(2 \log 2)]$ and an appropriate power of n , the envelope of $|u_n|$ has a bounded nonconstant modulation depending on the fractional part of $\log_2 n$. More precisely, along subsequences $\{\log_2 n\} \rightarrow \theta$, the normalized logarithm should admit a second-order limit depending periodically on θ .*

This is motivated by the exact dyadic shell factorization of $\hat{u}$, the Thue–Morse sign of its Fourier lobes, and the visible oscillations in figure 2. Proving it requires a uniform turning-point asymptotic for j_{2n} combined with the repository’s sharp lobe geometry.

12.2 Stable spectral biorthogonalization

The exact matrices $L^{(d)} A^{(d)} = I$ provide a natural starting point for preconditioning. The raw synthesis matrix contains factorially growing columns, but the dual kernel already incorporates the orthogonal projection that those columns are intended to invert.

Problem 12.3 (Optimal diagonal scaling of the spectral pair). *Find positive row and column weights that minimize*

$$\left\| D_1 A^{(d)} D_2 \right\|_2 \left\| D_2^{-1} L^{(d)} D_1^{-1} \right\|_2$$

subject to exact biorthogonality. Determine whether the optimal condition grows subfactorially, and compare literal, oversampled, and compressed atom dictionaries.

Problem 12.4 (A genuinely stable exact loop). *Construct a block dictionary or multiscale packet in which the partial Legendre loop remains exact but the total variation of each block is bounded by a fixed polynomial in its degree. The adjacent non-flattening theorem shows that no naive regrouping of literal atoms can achieve this; the atoms or analysis functionals must be changed.*

12.3 Structure of the rational spectral constants

Research question 12.5 (Closed forms for $C_{m,r}$ and $T_{m\ell r}^{(M)}$). *Can the clipped integrals (7.5) and finite sums (7.8) be reduced to terminating hypergeometric or Wigner-symbol expressions with transparent two-adic valuations? The unclipped product of Legendre polynomials is governed by Gaunt coefficients; clipping at an affine support edge introduces incomplete beta sums. Their combination with the Rvachev–Appell samples may conceal a simpler q -binomial structure.*

Conjecture 12.6 (Two-adic boundedness after canonical normalization). *For fixed m and ℓ , there is an explicit affine function $a(r)$ such that $2^{a(r)} T_{m\ell r}^{(2^\ell)}$ is a two-adic integer for every r , and the normalized residues form a 2-regular sequence in r .*

The dyadic clipping endpoints and the q -integer annihilator make 2-regularity a natural possibility. The exact CSV data are suitable for automated kernel searches in the sense of Allouche–Shallit [12].

12.4 Lambert W and asymptotic thresholds

The repository uses W_{-1} to invert the small-argument endpoint relation of the Fabius function and the principal branch W to locate thresholds at which factorial reciprocal-MGF coefficients dominate prescribed tolerances. In the present setting, the asymptotic $|Q_{2n}(0)|^{1/(2n)} \sim 2n/(\pi e)$ implies that the degree at which the central atom coefficient reaches size X is governed to first order by inverting

$$2n \log \left(\frac{2n}{\pi e} \right) \asymp \log X,$$

thus

$$n \asymp \frac{\log X}{2W(\log X/(\pi e))}. \quad (12.2)$$

A sharper threshold must combine this factorial growth with the log-square and dyadically modulated decay of u_n .

12.5 Formalization roadmap

The new results divide cleanly into finite algebra and analytic bridges.

1. Define $Q_\ell = \mathcal{D}_u P_\ell$ in the existing polynomial deconvolution API and prove (3.4) from the Jacobi derivative theorem.
2. Instantiate the existing synthesis theorem with the Legendre basis and package the common-scale matrix $A^{(d)}$.
3. Define Λ_m as a convolution, prove support, smoothness, parity, and origin values, then prove $L^{(d)} A^{(d)} = I$ by finite integration.

4. Formalize clipped intervals and the rational polynomial integral $C_{m,r}(c)$; derive the parity theorem and the spectral sum rule.
5. Reuse the formal Fourier product and spherical-Bessel transform to prove (5.6); then formalize the elementary Bessel and log-normal tail estimates in section 9.
6. Derive endpoint-flatness sum rules from the already formalized C^∞ support properties and Legendre endpoint derivatives.
7. For the arithmetic conjectures, first formalize theorem 10.1, then develop two-adic valuations of the exact moment and Legendre coefficient recurrences.

13 Conclusion

The Legendre specialization of exact up-atom polynomial synthesis is completely explicit. Moment deconvolution converts P_ℓ into a rational Bell–Bernoulli/Jacobi polynomial Q_ℓ ; dyadic samples of Q_ℓ are the coefficients of a finite literal-shift representation on $[-1, 1]$. The repository’s minimal local dictionary compresses the same polynomial to $\ell + 2$ dilated atoms.

Substitution into the canonical Fourier–Legendre expansion closes the loop exactly. The correct object is an absolutely convergent series of finite cancellation blocks. The factorial growth of deconvolution and the adjacent non-flattening theorem rule out a naive unconditional atom series.

The dual kernel Λ_m reveals more than the requested composition. It turns the shifted atom system into an exact spectral analysis/synthesis pair, with $L^{(d)} A^{(d)} = I$. Expanding shifted atoms locally produces infinite rational identities for the canonical coefficients, while endpoint flatness produces an independent hierarchy of weighted cancellation laws. The spherical-Bessel representation connects Legendre coefficients directly to the dyadic sinc-product envelope and yields a quantitative log-square decay theorem.

The remaining frontier is not existence but structure: sharp oscillatory asymptotics, stable preconditioning, two-adic integrality, and closed forms for the rational spectral constants. These questions now have exact finite objects, reproducible data, and a clear route toward Lean formalization.

A Exact formulae used by the experiment program

For reference, the reciprocal EGF recurrence is

$$\gamma_{2m} = - \sum_{r=1}^m \binom{2m}{2r} \mu_{2r} \gamma_{2m-2r}. \quad (\text{A.1})$$

If

$$P_\ell(x) = \sum_{j=0}^{\ell} p_{\ell,j} x^j,$$

then the exact monomial form of Q_ℓ is

$$Q_\ell(x) = \sum_{j=0}^{\ell} p_{\ell,j} \sum_{r=0}^{\lfloor j/2 \rfloor} \binom{j}{2r} \gamma_{2r} x^{j-2r}. \quad (\text{A.2})$$

For rational c , the code evaluates

$$C_{m,r}(c) = \frac{2m+1}{2} \int_{\max(-1, c-1)}^{\min(1, c+1)} P_m(x) P_{2r}(x-c) dx \quad (\text{A.3})$$

by exact polynomial multiplication and antiderivatives.

B Artifact inventory

The accompanying archive contains:

- `legendre_rvachev_closed_loop.tex` and the compiled PDF;
- `experiments.py`, with detailed comments and command-line options;
- `README.txt` and `MANIFEST.txt`;
- exact CSV data for moments, reciprocal coefficients, Legendre coefficients, spectral sum rules, and synthesis checks;
- a multiprecision CSV file for the coefficient-decay experiment;
- the generated TeX snippets and PNG figures used in the report.

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